

Sowrav Chowdhury

Dhaka, Bangladesh | mail.sowravchy@gmail.com | IELTS Academic 7.5 | BSc EEE, CGPA 3.77/4.00

Research focus: AI-enabled respiratory sound and biomedical signal analysis for low-cost clinical monitoring.

Research Profile

Electrical and Electronic Engineering graduate with research experience in digital stethoscope systems, pediatric lung sound collection, respiratory sound quality assessment, and 1D-CNN pneumonia diagnosis support. Current Augmedix role adds practical exposure to U.S. clinical documentation workflows, EHR data quality, provider preferences, KPI/SLA monitoring, SQL/reporting, Retool, ClickUp, and AI-assisted documentation operations.

Education

BSc in Electrical & Electronic Engineering, University of Liberal Arts Bangladesh, 2023 - CGPA 3.77/4.00, Cum Laude; major in Electronics.

A Level, Science, Saint Francis Xavier Greenherald International School, 2018; O Level, Science, 2015.

Research Experience

Research Assistant, University of Liberal Arts Bangladesh | Sep 2023-Jan 2025

- Worked on an ICT Innovation Fund project involving digital stethoscope development, pediatric respiratory sound collection, and healthcare signal data quality improvement.
- Contributed to a structured lung-sound database with 1,872 recordings from 117 pediatric subjects.
- Designed and tested 3D-printed hardware adapters for biomedical data collection systems.
- Coordinated hospital survey activity and collaborated with healthcare professionals to improve data reliability.

Clinical Data and Workflow Experience

Medical Data Processor / Squad Lead, Augmedix Bangladesh | Feb 2025-Present

- Lead squad-level clinical documentation workflows, KPI/SLA tracking, quality monitoring, and daily operational performance.
- Use SQL, PostgreSQL, MySQL, Excel, Google Sheets, Retool, and ClickUp for workflow status, documentation progress, follow-ups, and SLA monitoring.
- Review EHR-based clinical records for accuracy, completeness, consistency, and documentation quality.
- Support U.S. healthcare workflows, including Meditech 5 ICC and provider-specific documentation preferences.

Selected Publications

- S. Chowdhury, A. B. M. S. U. Doulah, and M. Rasheduzzaman, "Quality Assessment of Respiratory Sounds Extracted from Self-Assembled Digital Stethoscopes," ICAEEE 2022. <https://doi.org/10.1109/ICAEEE54957.2022.9836502>
- S. Chowdhury, A. B. M. S. U. Doulah, M. Rasheduzzaman, and T. S. Rafa, "Pediatric Pneumonia Diagnosis: Integration of a Self-Assembled Digital Stethoscope with Raspberry Pi and 1D CNN Model," iCACCESS 2024. <https://doi.org/10.1109/iCACCESS61735.2024.10499564>
- M. S. Rahman, S. Chowdhury, M. Rasheduzzaman, and A. B. M. S. U. Doulah, "Artificial Intelligence-Based Algorithms and Healthcare Applications of Respiratory Inductance Plethysmography: A Systematic Review," Algorithms, 2024. <https://doi.org/10.3390/a17060261>

Technical Skills

Biomedical signal processing; respiratory sound analysis; machine learning / 1D-CNN; Python; MATLAB; Arduino/Raspberry Pi; SQL; PostgreSQL; MySQL; Excel/Google Sheets; Retool; ClickUp; research documentation; clinical workflow data quality.

Teaching, Awards, and Leadership

- Teaching Assistant, ULAB EEE | Sep 2022-Sep 2023: supported labs in Electrical Circuits, Microprocessor and Embedded Systems, Control Systems, Digital Electronics, and VLSI.
- Champion, Project Showcasing - IEEE Robotics and Automation Society, ULAB Student Branch.
- Semi-Finalist, Hult Prize Global Accelerator; Champion, Hult Prize OnCampus.
- Vice-Chancellor Honors List; Cum Laude Graduate; Beast Award - Level 1: NESSIE, Augmedix Bangladesh.

Research Fit Profile: Sowrav Chowdhury

AI-enabled respiratory sound and biomedical signal analysis for low-cost clinical monitoring

Research Identity

I am an EEE-trained biomedical signal processing applicant focused on low-cost respiratory monitoring tools that can survive real clinical workflow constraints. My strongest research anchor is a digital stethoscope / pediatric lung sound project for pneumonia diagnosis support, strengthened by current U.S. clinical documentation workflow exposure at Augmedix.

Evidence Snapshot

Area	Evidence
Dataset	1,872 lung sound recordings from 117 pediatric subjects; hospital-based collection.
Methods	Respiratory sound quality assessment, signal processing, Raspberry Pi-based acquisition, 1D-CNN pneumonia diagnosis support.
Publications	Three linked publications across IEEE ICAEEE, IEEE iCACCESS, and MDPI Algorithms.
Clinical workflow	Augmedix squad lead / medical data role: EHR data quality, provider preferences, documentation KPIs, SQL/reporting, Retool, ClickUp.
Fit direction	Respiratory sound AI, biomedical signals, wearable/low-cost monitoring, clinical data workflow-aware digital health.

Research Questions I Can Build Toward

- How can low-cost digital auscultation systems collect reliable pediatric lung sounds in noisy and low-resource clinical settings?
- How can signal quality metrics and clinically interpretable ML improve respiratory sound AI before deployment?
- How can wearable/mobile respiratory monitoring and EHR workflow knowledge be connected into usable clinical decision-support tools?

Selected Linked Work

- digital stethoscope respiratory-sound quality: <https://doi.org/10.1109/ICAEEE54957.2022.9836502>
- pediatric pneumonia with Raspberry Pi and 1D-CNN: <https://doi.org/10.1109/iCACCESS61735.2024.10499564>
- AI and respiratory inductance plethysmography review: <https://doi.org/10.3390/a17060261>